Tarek Messaoudi

Full Stack Engineer, Medical Imaging

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Work Experience

Software Engineer — **2026 - Present**

Chipiron // Medtech building an ultra-low-field MRI scanner for breast imaging

🖥️ MRI Operator Web Application

- Built the app from scratch, including the exam workflow: drag-and-drop protocols, live scan progress, abort
- Built a JSON Schema-driven form engine for acquisition parameters, with server-side field validation

🫁 DICOM Imaging Integration

- Connected Orthanc PACS storage, a reverse proxy with authenticated DICOMweb and a Cornerstone.js viewer

🏥 RIS/PACS Integration

- Imported RIS orders as planned exams via DICOM Modality Worklist, with patient data carried into DICOM files
- Sent series to a PACS via C-STORE with per-series retry; added patient search and data erasure

🔐 Access Control & Quality

- Implemented authentication and role-based access control in a FastAPI service
- Introduced front-end testing from scratch (Vitest, MSW, Playwright) and OpenAPI-generated API types

Full Stack Engineer — **2020 - 2025**

Qynapse // A healthcare technology company specializing in neuroimaging and AI

🧠 Medical Imaging Web Platform Redesign

- Refactored the MRI viewer and admin panel, adding segmentation correction, lesion tracking and 3D rendering

🧬 Reporting & Domain Tools

- Rebuilt the report pipeline (Jinja2 + Puppeteer) with templates per language and disease (Dementia, MS, ...)
- Built a normative Python library for imaging metrics and a 3D mesh microservice for structural MRI

📦 Scalable Infrastructure & Microservices

- Contributed to converting the academic BrainVisa platform into a Kubernetes-based scalable pipeline
- Expanded the customer-deployed DIMSE agent with routing logic and detailed logging; built an HL7 v2 client

Full Stack Engineer — **2017 - 2020**

Alter Solutions // A consulting and IT services company

- Built web apps from scratch for IRT System X (research dashboard), Thalès (inventory), PSA (file parser)
- Refactored SFR Altice's customer-service CRM with a new architecture and Mail/SMS notifications

Technical Skills

Programming Languages: TypeScript, JavaScript, Python, Java
Frontend Development: React, Tailwind CSS, Remix
Backend Development: NodeJs, Deno, FastApi, Spring
Databases: PostgreSQL, SQLite, MongoDB, REDIS
Other Technologies: Docker, Kubernetes, Git, Linux, RabbitMQ

Education

IMT Lille-Douai — **Software engineer major** // incl. a semester at Chungnam National Univ., South Korea **2012 - 2017**

Projects & Rewards

LOM (GitHub) - *Browser strategy game*

"Entreprise du Futur" Reward (Site) - *By Thalès, from Challenge Entreprendre*